

AI for Mathematics

Session 1: From Counting to AI for Mathematics

DSA5210 – National University of Singapore

13 May 2026

A Problem Knuth Had Been Working On



Donald Knuth (b. 1938)

American computer scientist at Stanford; author of *The Art of Computer Programming* and creator of the TeX typesetting system.

In Claude's Cycles, Knuth describes a graph-decomposition problem on a Cayley digraph, an AI-assisted search process, and a later human proof.

- The paper is dated 28 February 2026 and revised 14 April 2026.
- The AI-produced construction handled odd values of m .
- The even case is discussed separately in the postscript.

Sources: Knuth, "Claude's Cycles", 2026; Knuth news page.

Claude's Cycles: What Happened

"Shock! Shock!"

-- Donald Knuth, ["Claude's Cycles"](#)

- Claude reformulated the problem in terms of a directed Cayley graph.
- The successful odd-m construction appeared after exploration 31.
- Knuth then wrote the mathematical proof.
- A public Lean 4 formalization later formalized the main theorem.

The episode records search, programs, a construction, a human proof, and a later Lean formalization.

Formalization link

[kim-em/KnuthClaudeLean](#)

- Lean 4 repository.
- Formalizes results from Knuth's note.
- Main theorem:
`ClaudesCycles.hamiltonian_arc_decomposition`.

This is a source link, not a diagram of an agent system.

Before Proof

Mathematics as record, measure, and calculation

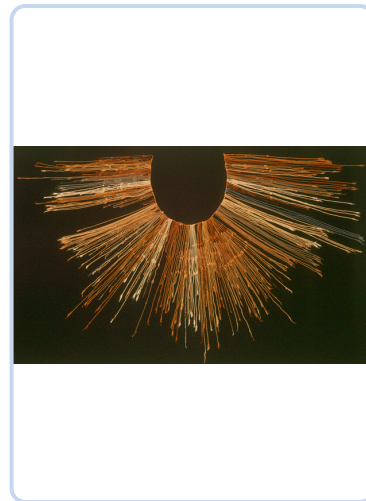
Counting Records

- The oldest mathematical traces are records: marks, knots, tokens, quantities.
- The [Ishango bone](#) is often discussed as an early tally artifact.
- [Quipu](#) records used knots and positions to encode information in the Andes.

At this stage mathematics is tied to memory: what has been counted, owned, owed, measured, or scheduled.



Tally artifact

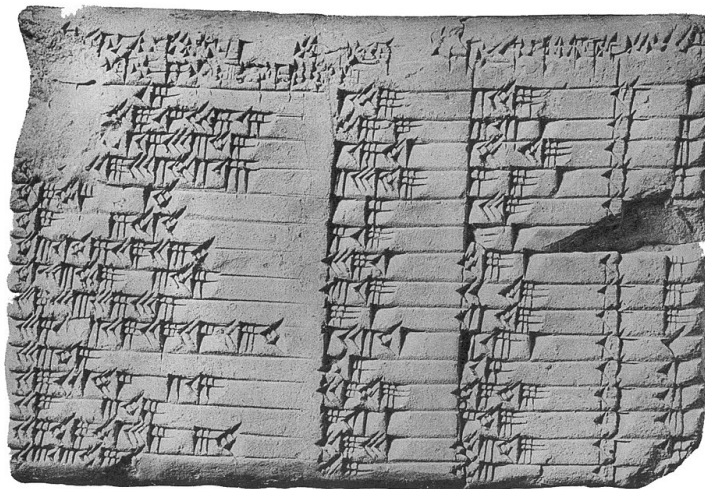


Quipu

Calculating Shape

- Ancient calculation handled land, building, astronomy, accounting, and calendars.
- [Plimpton 322](#) is a Babylonian tablet associated with Pythagorean triples.
- Such artifacts show structured computation before the Greek proof tradition.

$$3^2 + 4^2 = 5^2$$

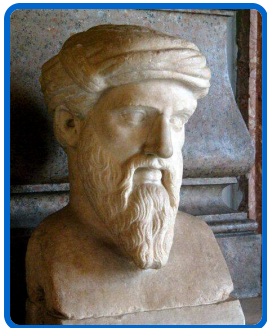


Babylonian mathematical tablet associated with Pythagorean triples.

Proof Appears

From examples to reasons

Pythagoras and $\sqrt{2}$



Pythagoras (c. 570-495 BCE)

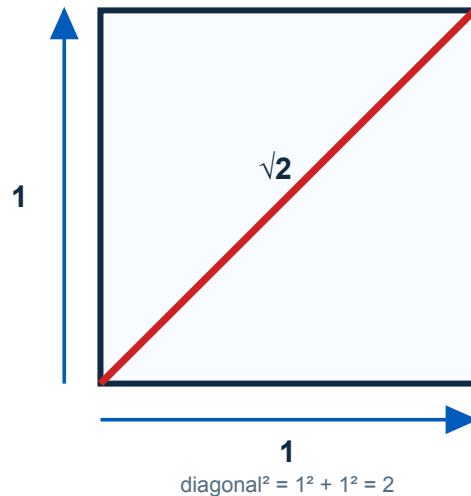
Greek philosopher and mathematician associated with the Pythagorean school and the theorem $a^2 + b^2 = c^2$.

$\sqrt{2}$ is irrational.

Hippasus legend: later sources connect disclosure of irrationality with drowning; this is not treated as established fact.

Sources: [Pythagoras](#); [square root of 2](#); SEP [Pythagoreanism](#); Britannica [Hippasus](#).

Unit Square



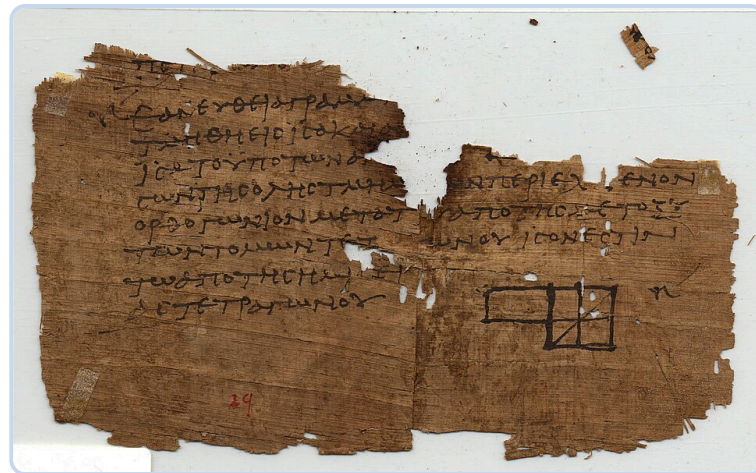
A unit square has diagonal length $\sqrt{2}$.

Euclid's *Elements*



Euclid of Alexandria (active c. 300 BCE)
Greek mathematician associated with *Elements*, a systematic presentation of geometry and number theory.

- Definitions, postulates, common notions.
- Propositions arranged so later claims depend on earlier proofs.
- A diagram may guide the eye; the proof carries the conclusion.



Fragment of Euclid's *Elements*.

Storing Mathematical Knowledge



[Al-Khwarizmi](#) (c. 780-850)

Mathematician and astronomer in Baghdad; his works are tied to the words algebra and algorithm.

9th c.

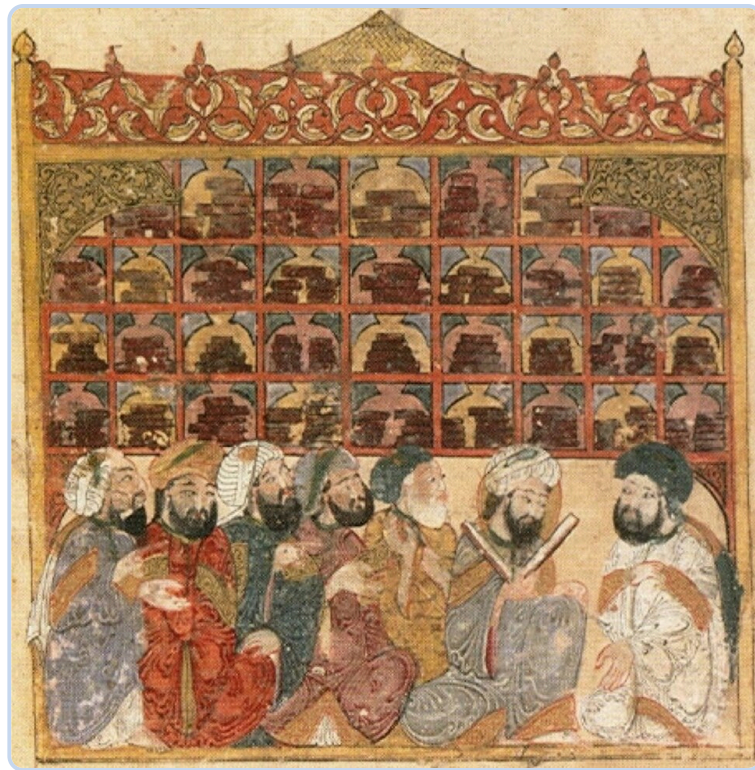
Translation and study around Baghdad's [House of Wisdom](#).

2005-

[Stacks Project](#) as an open, linkable reference for algebraic geometry.

2017-

[Mathlib](#) as a community library for formalized mathematics in Lean.



The House of Wisdom in Baghdad, represented in a historical illustration.

Proofs Become Hard To Check

Length, specialization, and verification

Fermat's Last Theorem: Statement and Gap



Pierre de Fermat (1607-1665)

French lawyer and mathematician whose margin note led to Fermat's Last Theorem.



Andrew Wiles (b. 1953)

British mathematician who proved Fermat's Last Theorem in the 1990s.

$a^n + b^n = c^n$, with $n > 2$, has no positive integer solutions.

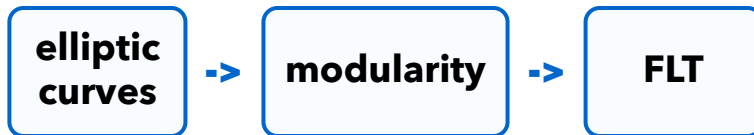
Fermat wrote his claim in 1637. Wiles announced a proof in 1993; after a gap was found, a corrected proof appeared in 1994 and was published in 1995.

Andrew Wiles: A Modern Proof



Andrew Wiles proved a deep case of the modularity theorem for elliptic curves, from which Fermat's Last Theorem follows.

- Modern proof can depend on a large body of specialized theory.
- Checking is distributed across experts and texts.
- A formal project must also formalize prerequisites.



This is the kind of proof where the statement is short but the verification context is large.

ABC Conjecture: Verification as Part of the Story



Shinichi Mochizuki (b. 1969)

Japanese mathematician at Kyoto University's RIMS, known for work in arithmetic geometry and Inter-universal Teichmuller theory.

- In 2012 he released papers claiming a proof of the ABC conjecture.
- The claimed proof uses Inter-universal Teichmuller theory.
- Acceptance and verification have been debated among experts.

For this course, ABC is a case where the verification process itself became part of the mathematical story.

Sources: Asahi/Kyoto University photo caption, [2021 article](#); [Plus Magazine overview](#); [ABC conjecture](#).

ABC: The Scholze-Stix Objection



Peter Scholze (b. 1987)

German arithmetic geometer and Fields Medalist; known for perfectoid spaces.



Jakob Stix (b. 1974)

German arithmetic geometer at Goethe University Frankfurt; works in anabelian geometry.

- In 2018 Scholze and Stix wrote [*Why abc is still a conjecture*](#).
- Their objection focuses on the step around IUT III, Corollary 3.12.
- The Mochizuki side disputed their interpretation and whether it constitutes a gap.

This is an example of mathematical verification becoming a public, technical disagreement.

Sources: Peter Scholze and Jakob Stix, [*Why abc is still a conjecture*](#), 2018; Quanta, [*Titans of Mathematics Clash Over Epic Proof of ABC Conjecture*](#), 2018; Stix photo: [Wikimedia Commons](#).

LANA: Lean Formalization of IUT

LANA stands for **Lean for ANAbelian geometry**.

- The project digitizes definitions and results from anabelian geometry using Lean4.
- ZMC announced LANA on 31 March 2026.
- The project says preparation began in autumn 2023 and full operation began in September 2024.

One aim is to formalize major theorems of anabelian geometry and build a Lean library.

Another aim is to verify Inter-universal Teichmüller theory through Lean formalization, from a neutral standpoint.

The target is not an AI opinion about ABC; it is a proof-assistant route to precise statements and checkable dependencies.

Formal Foundations and Proof Checking

Can checking become mechanical?

Hilbert's Program



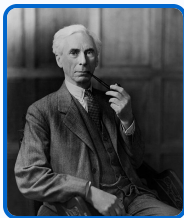
David Hilbert (1862-1943)

German mathematician whose work shaped algebra, geometry, analysis, foundations, and mathematical physics.

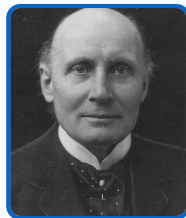
Hilbert's program aimed to place mathematics on explicit formal foundations and prove their consistency using finitary methods.

- Mathematical Problems (1902): the list of 23 Hilbert problems.
- Formal language.
- Explicit proof rules.
- Mechanical checking as an imagined endpoint.

Principia Mathematica



Bertrand Russell (1872-1970)
British philosopher, logician, and mathematician.



Alfred N. Whitehead (1861-1947)
British mathematician and philosopher.

Principia Mathematica (1910-1913) attempted to derive mathematics from logic through explicit symbolic rules.

PRINCIPIA MATHEMATICA

TO *56

BY
ALFRED NORTH WHITEHEAD
AND
BERTRAND RUSSELL, F.R.S.



CAMBRIDGE
AT THE UNIVERSITY PRESS

The famous $1 + 1 = 2$ result appears hundreds of pages into the work.

Gödel's Incompleteness Theorems



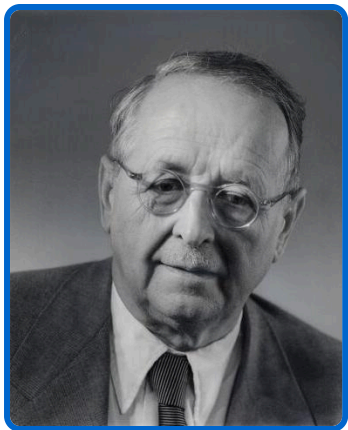
Kurt Gödel (1906-1978)

Austrian-American logician whose incompleteness theorems changed the foundations of mathematics.

For any consistent, effectively axiomatized formal system T that can express elementary arithmetic:

1. There is an arithmetic sentence φ_T such that T proves neither φ_T nor its negation $\neg\varphi_T$.
 2. T cannot prove its own consistency statement $\text{Con}(T)$.
- Gödel's incompleteness theorems limit Hilbert's hope for one complete formal foundation.
 - They do not prevent formal checking of particular proofs in a specified system.

Weyl: Computing Machines as Outside Help



Hermann Weyl (1885-1955)

German-American mathematician; worked in analysis, geometry, mathematical physics, topology, and foundations.

In 1946, Weyl publicly raised the possibility that high-speed computing machines could become outside help for increasingly difficult mathematical work.

- This is before modern proof assistants.
- The following examples involve search, checking, and computer-assisted calculation.

Sources: Hermann Weyl; Princeton Companion to Mathematics, article VI.80.

Automated Proof Search: Logic Theorist

1956

Logic Theorist searches for proofs in symbolic logic.

Input

Theorems from Chapter 2 of *Principia Mathematica*.

Result

Proofs for 38 of the first 52 target theorems.

This is the proof-search tradition: the machine tries to find a derivation from formal rules.

It is different from proof checking: a found proof still needs a trusted way to verify each step.

Source: History of Information, Logic Theorist.

Proof Checking: Automath

1968

[Automath](#) is designed as a language for fully written mathematical proofs.

Object

A proof becomes a precise formal object, not only a prose argument.

Checker

The computer verifies whether the proof object follows the rules.

Automath makes the checking problem explicit: if a proof is written in the formal language, a small checker can inspect it line by line.

```
ε * N := PN : set
ε * 0 := PN : N
ε * n := --- : N
n * s := PN : N
ε * 1 := s(0): N
```

`---` opens an assumption line. `PN` introduces a primitive notion. A definition line gives the body and the type.

Computer-Assisted Proof: Four Colors

1976

The [Four Color Theorem](#) is proved using computer-assisted case checking.

Issue

Part of the proof depends on computations too large to check by hand in the usual way.

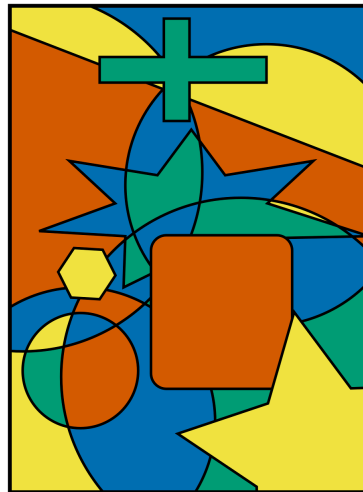
2005

A proof is formalized and mechanically checked in [Coq](#).



Georges Gonthier

Computer scientist; led the Coq formalization of the Four Color Theorem.



A planar map colored with four colors.

Sphere Packing and Flyspeck



Johannes Kepler (1571-1630)

German astronomer and mathematician; formulated the sphere-packing conjecture in 1611.



Thomas Hales (b. 1958)

American mathematician; proved the Kepler conjecture and led the Flyspeck formalization.

1611

The conjecture asks for the densest packing of equal spheres in three-dimensional Euclidean space.

1998

Hales announced a computer-assisted proof.

2014

The [Flyspeck project](#) completed a formal proof using HOL Light and Isabelle.

Optimal density: $\pi/\sqrt{18}$.

Proof Assistants Become Infrastructure

1970s

Mizar and HOL-family provers help establish reusable formal proof environments.

1980s

Coq and Isabelle develop proof languages, logics, automation, and library practices.

1994

The [QED Manifesto](#) proposes a computer-based body of formalized mathematical knowledge with mechanically checked proofs.

Library

Reusable definitions, libraries, notation, automation, and checkers become central.

Modern

Lean and Mathlib are a library-centered proof-assistant ecosystem.

Sources: [QED Manifesto](#), 1994; ["Twenty Years of the QED Manifesto"](#), 2016.

Lean, Mathlib, Formal Libraries

Proofs as checkable objects

Why Lean Was Built

Earlier systems

- Automath, Mizar, Coq, Isabelle, HOL-family systems.
- Each explored a different balance of logic, automation, libraries, and interaction.

Lean's design space

- Dependent type theory with a small trusted kernel.
- Automation and tactic programming.
- A programming language for metaprogramming.
- A large community library for mathematics.

The question was not simply "Can a machine check proofs?" It was how to make formal mathematics practical enough for a growing community.

Sources: [Lean official site](#); [Theorem Proving in Lean 4](#).

Leonardo de Moura and Lean



Leonardo de Moura

Computer scientist known for the Z3 SMT solver and as the creator of the Lean theorem prover.

- Lean began at Microsoft Research around 2013.
- Lean 4 is both a theorem prover and a programming language.
- The system supports formal proof, automation, and metaprogramming in one environment.

For AI-for-math, Lean provides a checker, a language, and a searchable mathematical codebase.

Sources: de Moura [homepage](#); [Lean official site](#); Microsoft Research [Z3 project](#).

What Lean Changes About Trust

Traditional checking

Reader checks the written argument.

Trust depends on expertise, exposition, and social review.

Long prerequisites may be implicit.

Lean checking

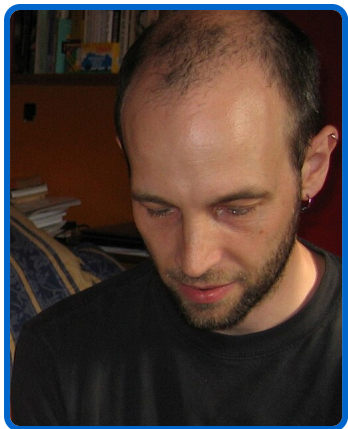
The kernel checks a formal proof term.

Checking is independent of the prover's identity.

The formal statement must match the intended theorem.

Careful wording: Lean does not make every written proof automatically correct. It checks a formalized theorem and proof inside a specified environment.

Kevin Buzzard, Xena, and Mathlib



Kevin Buzzard

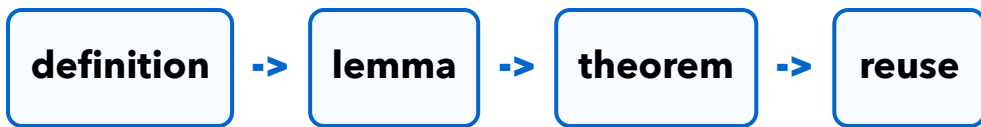
British number theorist at Imperial College London; an advocate for formalizing modern and undergraduate mathematics in Lean.

- The [Xena Project](#) used Lean to formalize undergraduate mathematics.
- Undergraduate curricula give controlled tests for definitions, APIs, and missing lemmas.
- Reusable results can become part of Mathlib.

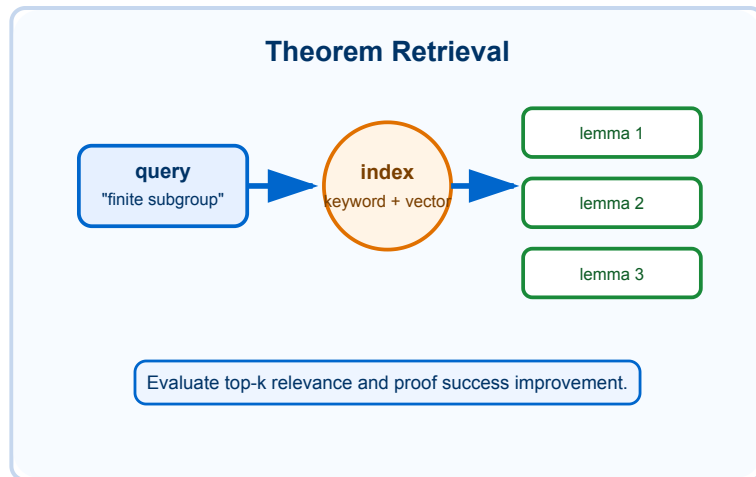
Mathlib did not simply start as one undergraduate project. But projects such as Xena helped expose what a real mathematical library needs.

What Mathlib Is

- [Mathlib](#) is Lean's community library of formalized mathematics.
- It contains definitions, theorems, tactics, notation, and documentation.
- It acts as mathematical memory for both humans and proof-search tools.



Sources: [Mathlib overview](#); The mathlib Community, "[The Lean Mathematical Library](#)", 2019.



Theorem retrieval: query, index, candidate lemmas, and evaluation.

Formalization Projects

2020

[Liquid Tensor Experiment](#) begins from a challenge to formalize sophisticated condensed mathematics.

2023

The [PFR formalization project](#) organizes public Lean formalization around a theorem in additive combinatorics.

2024-

The [FLT Project](#) works toward a Lean formalization of Fermat's Last Theorem.

2026

Knuth's [Claude cycles proof](#) is formalized soon after the written proof appears.

These projects are not just theorem statements. They require definitions, dependencies, library design, and coordination.

Formalization Time Scales

**2003-
2014**

[Flyspeck](#) gives a completed formal proof of the Kepler conjecture after a decade-scale project.

**2020-
2022**

[Liquid Tensor](#) verifies a main theorem from Clausen-Scholze mathematics in Lean.

2023

[PFR](#) develops a public Lean formalization around a new additive-combinatorics proof.

2025

[Gauss](#) reports an AI-assisted Lean formalization of the strong Prime Number Theorem in three weeks.

Compare cautiously: scope, starting library, automation, and human coordination differ from project to project.

Tao-Style Public Formalization



[Terence Tao](#) (b. 1975)

Australian-American mathematician, Fields Medalist, known for work across analysis, number theory, combinatorics, and mathematical exposition.

- The [PFR project page](#) lists formalization goals, dependency structure, and progress.
- Tasks can be split across contributors.
- Lean/Mathlib gives a shared checker for contributed pieces.

This is a public development style: theorem, blueprint, dependency list, library PRs, checked proof.

Peter Scholze and Liquid Tensor



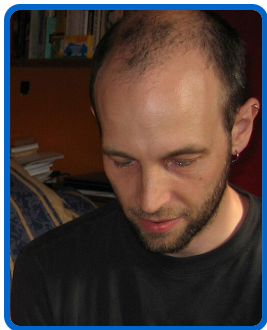
[Peter Scholze](#) (b. 1987)

German mathematician, Fields Medalist, known for perfectoid spaces and arithmetic geometry.

- Scholze posed a formalization challenge connected to condensed mathematics.
- The [Liquid Tensor Experiment](#) became a public Lean formalization project.
- The project tested whether Lean/Mathlib could handle very recent research-level structures.

Formalization here is also library construction: new definitions must be reusable enough for later mathematics.

FLT Project



Kevin Buzzard's [FLT Project](#) aims to formalize the proof of Fermat's Last Theorem in Lean.

- The short theorem statement hides large prerequisite towers in modern number theory.
- The project page tracks missing Lean infrastructure and milestones.
- The intended endpoint is a checker-verified Lean proof.



The formalization task includes building enough reusable library mathematics for Wiles's proof to type-check.

Source: Imperial College London, [FLT Project](#).

AI Enters Mathematical Reasoning

Generation, search, tools, and verification

What Benchmarks Measure

Benchmark type	Typical output
Answer benchmarks	number, expression, multiple choice
Competition proof	natural-language proof or formal code
Formal proving	Lean theorem and proof script
Research frontier	new proof attempt, construction, or verified answer

Read each score with its checker

- Answer key, symbolic checker, or Python verifier.
- Human grading of a written proof.
- Lean kernel checking a formal proof.
- Expert review for research-level claims.

Recent benchmark history records rapid improvement in mathematical problem-solving, but the artifact being scored changes across benchmarks.

From Benchmarks to Research Attempts

2024

[FrontierMath](#) is introduced; its paper reports state-of-the-art models solving under 2% at release.

2025

IMO announcements report gold-medal-level systems, with natural-language, formal, and hybrid routes discussed below.

2026

[FrontierMath: Open Problems](#) tests proposed solutions to open research problems with programmatic verifiers.

2026

[First Proof](#) asks for checkable proof attempts on ten domain-specific research-level problems.

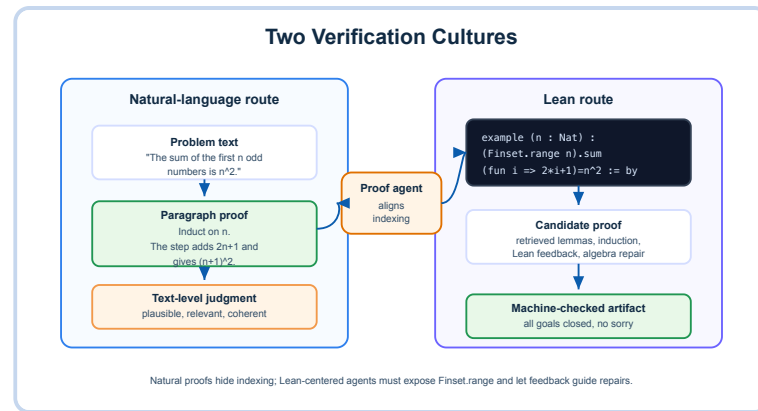
Epoch's 11 May 2026 FrontierMath update says an AI-assisted review flagged fatal errors in about one third of Tiers 1-4 problems.

Sources: Glazer et al., [FrontierMath](#), 2024; Epoch AI [Tiers 1-4 update](#), 2026; Epoch AI [Open Problems](#), 2026; OpenAI [First Proof submissions](#), 2026.

IMO Grand Challenge

- The [IMO Grand Challenge](#) proposed a benchmark for AI systems solving International Mathematical Olympiad problems.
- Each IMO problem is worth 7 points, for a total of 42.
- Modern systems differ in what they output and how that output is checked.

Read each announcement by asking: What was generated? Who or what checked it? Was the proof natural language, formal code, or a hybrid?



Natural-language proof route and formal-proof route.

IMO 2024: Formal and Neuro-Symbolic Route

**Jan
2024**

[AlphaGeometry](#) combines a language model that proposes auxiliary constructions with a symbolic geometry engine.

**Jul
2024**

[AlphaProof + AlphaGeometry 2](#) solve IMO problems at silver-medal standard.

Output

AlphaProof works with formal proof code; AlphaGeometry uses a symbolic checker in the geometry component.

Interface

Problems and proofs require translation between natural language and formal representations.

Source: Google DeepMind, "[AI achieves silver-medal standard solving International Mathematical Olympiad problems](#)", 2024.

AlphaProof: Lean Output Fragment

- DeepMind released Lean files for IMO 2024 problems P1, P2, and P6.
- The README says `originals/` contains direct agent output with explanatory comments added.
- Edited versions compile on a public Lean/mathlib release.

The excerpt contains anonymous variables, placeholder goals, and dense tactic chains.

```
-- IMO 2024 P1 excerpt, not a standalone file
existsλx L⇒(L 2 two_pos).rec λl Y⇒?_
useλy . x⇒y.rec λS p⇒?_
· simp_all[λL:ℕ⇒(by norm_num[Int.floor_eq_iff]:
  [(L:ℝ)*S]=L*S)]
  rw[p.2, Int.dvd_iff_emod_eq_zero,
    Nat.lt_iff_add_one_le, ←Finset.sum_mul,
    ←Nat.cast_sum, S.even_iff] at *
  simp_all[Finset.sum_range_id]
```

Source: Google DeepMind, [AlphaProof outputs from the 2024 IMO](#) public Lean package.

IMO 2025: Natural-Language Route

OpenAI announcement

- Experimental reasoning LLM.
- Natural-language proofs.
- Reported score: 35/42.
- Publicly announced by OpenAI researchers; not an official IMO entry.

Source: [Techmeme archive of the OpenAI announcement](#)

Gemini Deep Think

- Advanced Gemini model with Deep Think.
- End-to-end natural-language proofs from official problem descriptions.
- Officially achieved gold-medal standard.
- Google describes parallel, extended reasoning behavior.

Source: Google DeepMind, [official IMO 2025 post](#)

IMO 2025: Formal-Proof Route

Harmonic Aristotle

- Lean-oriented automated theorem-proving system.
- Reports IMO-level automated proving with formal proofs.
- Architecture emphasizes decomposition, search, and verification.

Source: Harmonic Team, ["Aristotle: IMO-level Automated Theorem Proving"](#).

Seed-Prover / Seed-Prover 1.5

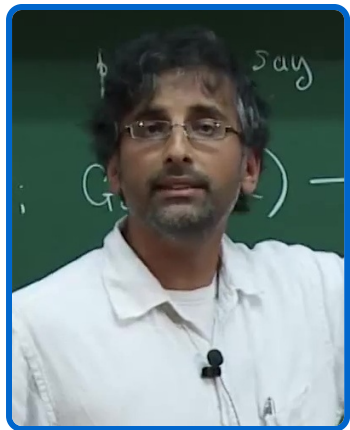
- Lean proof generation with agentic reinforcement learning and tool use.
- Seed-Prover 1.0 participated in IMO 2025.
- Seed-Prover 1.5 later reports Lean proof code for five IMO 2025 problems.

Source: ByteDance Seed, [Seed-Prover 1.5 post](#); [GitHub](#); [Seed-Prover paper](#).

Mathematicians on AI

Concrete public positions and working habits

Akshay Venkatesh



Akshay Venkatesh (b. 1981)

Australian mathematician, Fields Medalist, known for work connecting number theory, dynamics, topology, and representation theory.

- His public essays and interviews discuss how AI may change mathematical practice.
- One recurring theme is that AI can alter the roles of exploration, criticism, and exposition.
- For this course, his perspective belongs to the question of workflow, not only theorem proving.

Source: Institute for Advanced Study, ["The AI Revolution in Math Has Arrived"](#); Quanta, [2026 interview/essay package](#).

Timothy Gowers



Timothy Gowers (b. 1963)

British mathematician, Fields Medalist, known for functional analysis, combinatorics, and the Polymath collaborative projects.

- On 8 May 2026 he posted [a detailed account](#) of working with ChatGPT 5.5 Pro on a research problem.
- He describes the interaction as research collaboration rather than a finished formal verification pipeline.
- The post is useful here because it records a mathematician's working process with the model.

"PhD-level research in an hour or so" -- Gowers, 2026.

Terence Tao



Terence Tao appears twice in today's story: as a research mathematician using public collaboration, and as a public commentator on AI-math claims.

- His PFR formalization page shows a public, dependency-driven proof-development workflow.
- His commentary on AI-math announcements often distinguishes natural-language proof, formal proof, and human verification.
- That distinction is the same one used in the IMO architecture slides.

Sources: PFR formalization project; Tao's public blog.

Ken Ono



Ken Ono (b. 1968)

American number theorist known for work in modular forms, partitions, and arithmetic.

- Ono's recent public role connects mathematics, institutions, and AI-for-math development.
- His example is about building research environments around AI tools, not only using one model in isolation.
- For student projects, this points to systems: data, tools, verification, and collaboration.

Sources: University of Virginia [faculty profile](#); public AI-for-math reporting linked in lecture notes.

Assessment

Assignments

Component	Weight
Assignment 1: AI-assisted natural-language proof	20%
Assignment 2: AI-assisted Lean formalization of A1	20%
Assignment 3: Group AI-for-math project	20%
Individual report: submitted by each student	40%

Assignment 3

- Teams of 3 students maintain a shared Git repository.
- Possible forms: agent design, theorem discovery, or mathematical problem solving.
- Examples: proof critique, Lean assistance, conjecture search, PDE problem solving.
- Individual grades may reflect contribution amount and quality.

Each artifact records the mathematical target, AI workflow, checking method, and failure analysis.

Open Problem for Live AI Dialogue

Hadwiger--Nelson problem

Color every point of $\mathbb{R}^2$ so that points at distance **1** have different colors.

What is the minimum number of colors?

Known range:

$$5 \leq \chi(\mathbb{R}^2) \leq 7.$$

Ask AI: Is the problem solved?

Dialogue goals

Separate a solved theorem from an open problem.

Ask for definitions: unit-distance graph, chromatic number, finite witness.

Ask which claims are mathematical, computational, or still open.

After Class: Try an AI Coding Agent

Setup

- Register or verify a GitHub account.
- Set up Codex or Claude Code on your machine.
- Create or pull a repository via Codex.

What to try

- Ask it to analyze a Git repository: files, scripts, dependencies, build commands.
- Ask it to help with a math problem: assumptions, examples, possible proof routes.
- Ask it to review a proof draft: gaps, missing definitions, unsupported steps.
- For harder reasoning tasks, try `xhigh` mode and compare with a faster mode.

Keep a short note: what you asked, what it produced, and what you checked.